Experiment (2)
Simulation of Diode Circuits using LT Spice

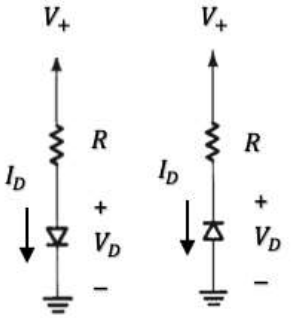
Objectives

- To study the IV characteristics of a standard diode (1N4001) by simulation.
- To simulate simple DC and AC diode circuits.

Pre-lab Work

Analyze the circuits below using the constant voltage diode model (V_{DO}=0.7 V).

Circuits



Forward Backward

Figure 1: Diode IV Circuit.

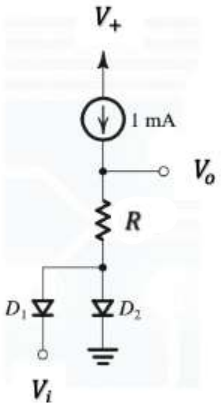


Figure 2: DC diode Circuit.

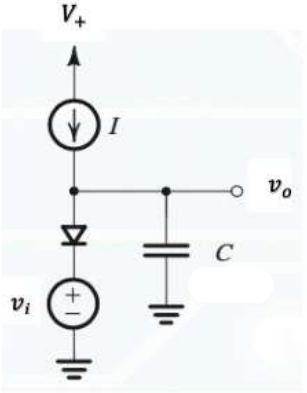


Figure 3: AC Diode Circuit

Procedure

Part (1): Diode IV Characteristics.

1. Connect the circuit in figure 1 (Forward). Use the circuit elements values given by the lab instructor. V_+ is a DC voltage source.
2. Use the DC Sweep function and simulate the response of the circuit to produce the IV curve of the diode (I_D vs. V_D). What is the V_{DO} of this diode? Record the results in table 1.
3. Repeat for the backward circuit. What is the V_{ZO} of this diode? Record the results in table 1.
4. Vary the value of R and repeat the procedure observing the effects (if any).

Part (2): DC Diode Circuit.

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. V_+ and V_i are DC voltage sources.
2. Simulate the circuit. Which diode is ON and which is OFF?

3. With V_i fixed, vary the value of R and repeat step 2 observing the effects on the diodes switching. Record your results in table 2.
4. With R fixed, vary V_i between the range -10 to +10 V and plot the output characteristics curve (V_o vs. V_i).

Part (3): AC Diode Circuit.

1. Connect the circuit in figure 3. Use the circuit elements values given by the lab instructor. I in the circuit is a DC current source and v_i is a sinusoidal signal with small amplitude (less than 10 mV) and a frequency f.
2. Simulate the circuit and plot the waveforms for v_i and v_o.
3. From the plot in step 2, find the phase shift between v_i and v_o. Record the result in table 3.
4. Vary I and observe the phase shift between v_i and v_o at each setting. Record your results in table 3. Find a value of I that gives a phase shift of $\pm 45^o$?

Results and Analysis

Part (1): Diode IV Characteristics.

Table 1. Diode IV Curve.

R (kΩ)	Simulation V_{DO} (V)	Data sheet V_{DO} (V)	Simulation V_{ZO} (V)	Data sheet V_{ZO} (V)

1. Comment on the results in table 1.

2. Do the IV curves that you obtained resemble the standard diode characteristics? Explain.

Part (2): DC Diode Circuit.

Table 2. Simulated DC results.

R (kΩ)	Diode 1 (ON/OFF)	Diode 2 (ON/OFF)	V_o (V)

1. Comment on the results in table 2.

2. Comment of the output characteristics curve.

Part (3): AC Diode Circuit.

Table 3. Simulated AC results.

I (mA)	Phase shift

1. Which value of I gives a phase shift of $\pm 45^{o}$?

2. Explain the output waveform.

3. Comment on the circuit output characteristics curve.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							